

Report 08 — Detection Results

Project: SIEM Correlation Rules — ELK Stack **Date:** 2026-07-07 | **Log Date:** 2026-06-15 **Version:** 1.0

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1. Executive Summary of Results

Across the 5.5-hour monitoring window on 2026-06-15, the 13 custom correlation rules generated a total of ****150 alert events**** across all three attack categories. All three attack campaigns were successfully detected. No confirmed attack activity was missed at the technique level given the available log sources.

Category	Rules Deployed	Total Alerts	True Positives	False Positives	Detection Rate
Credential Stuffing	4	~8	8	0	100%
DNS Tunnelling	4	~107	107	0	100%
PowerShell Exploitation	5	~33	33	0	100%
Total	13	~148	~148	0	100%

The zero false-positive count reflects the specificity of the rules as tuned against this dataset. Real-world deployment would require further baselining to maintain this precision.

2. Alert Results by Rule

2.1 Credential Stuffing Rules

Rule ID	Name	Alerts Fired	TP	FP	First Alert	Last Alert
CS-001	SSH Credential Stuffing	3	3	0	00:43:39	03:08:27
CS-002	Windows Logon Failure Storm	2	2	0	01:00 (est.)	02:33 (est.)
CS-003	Account Lockout Storm	1	1	0	01:37:13	01:37:13
CS-004	Success After Failures	2	2	0	01:59 (est.)	03:28:22

CS-001 Alert Details:

Alert #	Source IP	Targeted Accounts	Failure Count	Window
1	185.220.101.47	ubuntu, backup, deploy, oracle	4+ failures	00:28–00:48
2	185.220.101.47	admin, administrator, root, deploy	4+ failures	01:01–01:22
3	185.220.101.47	postgres, svc_backup, administrator, deploy	4+ failures	02:22–03:08

CS-003 Alert Detail:

Trigger time: 2026-06-15 01:37:13
Threshold met: 3 distinct lockouts within 30 minutes
Locked accounts: kpatel (01:02), nreddy (01:25), rgupta (01:37)
Subsequent: lchen (01:56), dwilson (01:59) – escalation confirmed
Severity: CRITICAL

2.2 DNS Tunnelling Rules

Rule ID	Name	Alerts Fired	TP	FP	First Alert	Last Alert
DNT-001	Excessive TXT Volume	~8	8	0	00:52 (est.)	01:54 (est.)

Rule ID	Name	Alerts Fired	TP	FP	First Alert	Last Alert
DNT-002	High-Entropy Subdomain	~47	47	0	00:10:09	01:54:50
DNT-003	Low TTL TXT Response	~47	47	0	Zeek data	Zeek data
DNT-004	DoH POST Unapproved	4	4	0	00:32:34	00:55:45

DNT-004 Alert Details:

Alert #	Timestamp	User	Source IP	DoH URL	Response
1	00:32:34	lchen	10.20.4.97	https://cdn-sync-update.net/dns-query	200 (897 bytes)
2	00:33:22	svc_sql	10.20.4.19	https://cdn-sync-update.net/dns-query	200 (435 bytes)
3	00:47:45	nreddy	10.22.9.33	https://cdn-sync-update.net/dns-query	200 (281 bytes)
4	00:55:45	kpatel	172.16.5.128	https://cdn-sync-update.net/dns-query	200 (842 bytes)

2.3 PowerShell Exploitation Rules

Rule ID	Name	Alerts Fired	TP	FP	First Alert	Last Alert
PS-001	Encoded PS with Hidden Window	6	6	0	02:22:26	05:04:26
PS-002	Download Cradle	10	10	0	00:25:46*	05:10:07
PS-003	Office Spawning PowerShell	10	10	0	00:46:06	01:55:48
PS-004	regsvr32 Remote Scriptlet	4	4	0	00:44:25	02:15:05
PS-005	rundll32	3	3	0	00:45:26	01:57:51

Rule ID	Name	Alerts Fired	TP	FP	First Alert	Last Alert
	from Temp					

*PS-002 counts proxy log evidence (stage2.ps1 downloads) starting 00:25:46, plus PowerShell Event 4104 captures starting 02:07:56.

PS-002 High-Priority Alert Breakdown:

Timestamp	Host	User	Type	Payload
00:25:46	(proxy)	svc_backup	DownloadString	stage2.ps1
00:41:18	(proxy)	jsmith	DownloadString	stage2.ps1
01:20:09	(proxy)	rgupta	DownloadString	stage2.ps1
02:07:56	WKS-SALES22	rgupta	IEX DownloadString	stage2.ps1
02:55:39	SRV-APP05	svc_backup	DownloadFile	update.exe from 185.220.101.47
03:29:53	SRV-APP05	nreddy	Encoded -Enc	Embedded WebClient
04:20:07	WKS-SALES22	svc_backup	DownloadFile	update.exe from 185.220.101.47
04:51:46	SRV-DC01	jsmith	IEX DownloadString	stage2.ps1 ← DC Compromise
04:59:36	SRV-DC01	nreddy	IEX DownloadString	stage2.ps1 ← DC Compromise
05:10:07	WKS-FIN07	kpatel	DownloadFile	update.exe from 185.220.101.47

3. Triggered Alert Timeline

The chronological sequence of alert triggers across all rules:

Timestamp	Rule	Severity	Host	User/IP	Summary
00:10:09	DNT-002	High	dns-srv01	10.22.9.19	High-entropy TXT query begins
00:25:46	PS-002	Critical	proxy	svc_backup	First stage2.ps1 download
00:32:34	DNT-004	High	proxy	lchen	DoH POST to cdn-sync-update.net
00:41:18	PS-002	Critical	proxy	jsmith	stage2.ps1

Timestamp	Rule	Severity	Host	User/IP	Summary
					download
00:43:39	CS-001	High	app-prod03	185.220.101.47	SSH brute force burst #1
00:44:25	PS-004	Critical	SRV-APP05	amehta	regsvr32 scriptlet
00:45:26	PS-005	High	WKS-SALES22	kpatel	rundll32 payload.dll
00:46:06	PS-003	Critical	SRV-APP05	kpatel	EXCEL spawning powershell
00:51:20	PS-003	Critical	SRV-FILE02	nreddy	EXCEL spawning powershell
00:54:10	PS-003	Critical	WKS-ENG14	svc_sql	WINWORD spawning powershell
00:55:45	DNT-004	High	proxy	kpatel	DoH POST to cdn-sync-update.net
01:01 (est.)	CS-002	High	multiple	various	Windows logon failure storm
01:02:15	CS-003 (partial)	—	WKS-FIN07	kpatel	First lockout (threshold not met yet)
01:20:09	PS-002	Critical	proxy	rgupta	stage2.ps1 download
01:25:38	CS-003 (partial)	—	WKS-ENG14	nreddy	Second lockout
01:37:13	CS-003	Critical	WKS-ENG14	rgupta	Lockout storm threshold MET
01:40:43	PS-004	Critical	SRV-APP05	amehta	regsvr32 scriptlet repeat
01:45:03	PS-004	Critical	SRV-FILE02	rgupta	regsvr32 scriptlet
01:55:48	PS-003	Critical	WKS-FIN07	tverma	cmd spawning powershell
01:57:51	PS-005	High	WKS-ENG14	rgupta	rundll32 payload.dll
02:07:56	PS-002	Critical	WKS-SALES22	rgupta	IEX DownloadStrin

Timestamp	Rule	Severity	Host	User/IP	Summary
					g — 4104
02:15:05	PS-004	Critical	WKS-SALES22	jsmith	regsvr32 scriptlet
02:22:26	PS-001	High	WKS-FIN07	tverma	Encoded PS hidden window
02:55:39	PS-002	Critical	SRV-APP05	svc_backup	DownloadFile update.exe
03:29:53	PS-001	High	SRV-APP05	nreddy	Encoded PS hidden window
04:20:07	PS-002	Critical	WKS-SALES22	svc_backup	DownloadFile update.exe
04:27:36	PS-001	High	WKS-FIN07	svc_sql	Encoded PS hidden window
04:48:27	PS-001	High	WKS-SALES22	kpatel	Encoded PS hidden window
04:51:46	PS-002	Critical	SRV-DC01	jsmith	IEX on Domain Controller
04:59:36	PS-002	Critical	SRV-DC01	nreddy	IEX on Domain Controller
05:04:26	PS-001	High	SRV-FILE02	dwilson	Encoded PS hidden window
05:10:07	PS-002	Critical	WKS-FIN07	kpatel	DownloadFile update.exe

4. Detection Accuracy

4.1 Metrics Summary

Metric	Value	Formula
True Positives (TP)	~148	Confirmed malicious alerts
False Positives (FP)	0	Alerts on benign activity
False Negatives (FN)	~3–5 (estimated)	Missed attacks (see Section 6)
True Negatives (TN)	(all non-alerting benign events)	—
Precision	100%	$TP / (TP + FP)$
Recall	~97%	$TP / (TP + FN)$
F1 Score	~0.985	$2 \times (P \times R) / (P + R)$

4.2 Detection Rate by Category

Credential Stuffing:		100%
DNS Tunnelling:		100%
PowerShell Exploit:		100%
Overall:		~97% (accounting for estimated FNs)

5. False Positive Analysis

5.1 Potential False Positive Scenarios

No confirmed false positives were triggered in this dataset. The following scenarios were evaluated for FP potential:

Scenario	Rule	Assessment	Mitigation
Get-ADUser export by kpatel on SRV-DC01 (00:45:05)	PS-002	Not FP — kpatel is not an AD admin; running this from SRV-DC01 is unauthorized	Restrict AD read access; flag non-admin Get-ADUser
svc_backup sudo nginx restart (00:53:46)	None triggered	True Negative — legitimate admin action; no rule covers sudo alone	Correct — sudo logging is informational only
jsmith accepted password SSH to mail-relay01 (00:33:49)	None triggered	True Negative — no prior failures from this source	Correct — no stuffing pattern for this session
Normal CRON jobs in auth.log	None triggered	True Negative — CRON runs are excluded from authentication rules	Correct
Legitimate DNS queries to office365.com, slack.com	DNT-001, DNT-002	True Negative — rules correctly exclude known-safe domains	Correct — allowlist working
Start-Process -WindowStyle Hidden (normal admin)	PS-001 (potential)	Low FP risk — -W Hidden alone is common; combined with -Enc is the trigger	The combined condition reduces FP

5.2 Tuning Recommendations to Maintain Low FP Rate

Rule	Tuning Action
CS-001	Add exception for known automated monitoring tools (Nagios, Zabbix) source IPs
CS-002	Suppress alerts for accounts with

Rule	Tuning Action
	PasswordNeverExpires service accounts during known batch windows
DNT-001	Expand safe domain allowlist as new SaaS tools are onboarded
PS-002	Add internal software distribution URLs (SCCM, Intune) to the exclusion list
PS-003	Add exception for signed, IT-managed macro-enabled templates with known hashes

6. False Negative Analysis

6.1 Estimated Missed Detections

Activity	Why Missed	Impact	Coverage Gap
Encrypted C2 over 443 to 91.203.145.12	No SSL inspection; firewall allows ESTABLISHED traffic outbound	High — persistent C2 not detected in real time	Deploy SSL/TLS proxy inspection
Potential LSASS dump by compromised accounts	No memory-level monitoring in log sources	Critical — credential theft from DC possible	Deploy EDR with LSASS guard
Persistence mechanisms (scheduled tasks, registry)	No Sysmon Event 12/13/14 or Event ID 4698 in dataset	High — attacker persistence would survive remediation	Enable Sysmon registry + file events
Lateral movement via SMB (internal)	network_connections.log shows internal 445 connections, but no content inspection	Medium — SMB-based lateral movement not correlated	Enable Windows Event ID 5140/5145
DGA-based C2 (if any beyond cdn-sync-update.net)	Only one tunnelling domain observed; others may exist	Medium — depends on attacker infrastructure breadth	Implement domain entropy ML model

6.2 Log Source Gaps

Missing Log Source	Impact on Detection
Sysmon Event IDs 7 (DLL Load), 10 (Process Access), 11 (File Create)	Cannot detect DLL injection, LSASS access, file drops
Windows Event ID 4698 (Scheduled Task Created)	Cannot detect persistence via scheduled tasks
Windows Event ID 4732/4728 (Group membership changes)	Cannot detect privilege escalation via group manipulation

Missing Log Source	Impact on Detection
NetFlow / IPFIX full telemetry	Cannot correlate byte volumes to specific user sessions
DNS response content from BIND (only Zeek has answers)	BIND logs only query direction; response content requires Zeek

7. Overall Findings

7.1 Campaign Summary

The log evidence from 2026-06-15 reveals a **coordinated, multi-stage attack campaign** targeting the CORP domain:

1. **Stage 1 — Reconnaissance & Initial Access (00:00–00:46):** SSH brute force from 185.220.101.47 targeting Linux servers. Macro-enabled Office documents distributed (inferred). First PowerShell stagers execute from EXCEL.EXE and WINWORD.EXE.
2. **Stage 2 — Execution & C2 Establishment (00:25–01:20):** Stage 2 PS1 payload downloaded from malicious-update-cdn.net by svc_backup, jsmith, and rgupta. DNS tunnelling begins to cdn-sync-update.net. DoH POSTs to the same domain establish a secondary covert channel.
3. **Stage 3 — Credential Harvesting (01:00–02:00):** Windows credential stuffing storm generates five account lockouts. LOLBAS techniques (regsvr32, rundll32) execute across SRV-APP05, SRV-FILE02, WKS-SALES22, WKS-ENG14.
4. **Stage 4 — Lateral Movement & Persistence (02:00–04:00):** Encoded PowerShell stagers spread across additional hosts. svc_backup downloads update.exe binary from 185.220.101.47 to AppData\Temp on two different hosts.
5. **Stage 5 — Domain Controller Compromise (04:51–05:10):** IEX download cradles execute on SRV-DC01 under jsmith (04:51) and nreddy (04:59). This represents full Active Directory compromise. Further update.exe downloads continue on WKS-FIN07.

7.2 Key Findings Table

Finding	Severity	Status
185.220.101.47 conducting sustained SSH brute force	High	Detected — CS-001
Five domain accounts locked out in 57 minutes	Critical	Detected — CS-003
DNS tunnelling via cdn-sync-update.net across 30+ hosts	High	Detected — DNT-001/002
PowerShell download cradle on SRV-DC01 (domain controller)	Critical	Detected — PS-002
svc_backup service account	Critical	Detected — PS-002

Finding	Severity	Status
downloading malware binary		
Office macros spawning encoded PowerShell across 10 host/user pairs	Critical	Detected — PS-003
regsvr32 LOLBAS loading remote scriptlet from cdn-sync-update.net	Critical	Detected — PS-004
Persistent ESTABLISHED C2 connections to 103.68.22.9, 91.203.145.12	High	Partial — no rule for encrypted C2